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International AS and A-level

Biology (9610)
BL03 Populations and genes
Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL BIOLOGY (9610) BL03 JUNE 2024

Good attempts were seen on all questions on the paper with some students demonstrating excellent subject knowledge.
It was good to see fewer generic responses to evaluation questions (**05.8** and **06.5**); students were obviously taking time to review methods and results rather than just listing phrases such as 'no statistical test'.
Application of knowledge questions, such as question 3, may require students to refer to a diagram to demonstrate their understanding of a process. The instruction in a question to **use information** from a diagram should direct students look at and understand the diagram before answering the question.
When answering genetic cross questions (**04.1** and **04.3**), students should make sure that the offspring genotype is matched to its phenotype.
The graph question 05.4 was answered better than in previous exam series; students were more likely to draw the correct type of graph and choose a suitable scale. Students should be reminded that a line of best fit can be a **curved** line.

QUESTION 01

01.1

Although good responses were seen, there were many answers referring to there being more food available for the mealybugs, or just stating that the mealybug were imported with the crop. Answers needed to focus on the information given in the question asking why the mealybugs have spread much **more** quickly in Africa and Asia.

01.2

Some very good answers were seen with over a third of students gaining 3 or 4 marks for identifying advantages and disadvantages of using parasitic wasps as a method of biological pest control. All marking points were seen with the most common advantage of biological control being its specificity, and the most common disadvantage was it becoming a pest itself.

Occasional misconceptions were seen with students describing pesticides leaching out of the soil and causing eutrophication, and a number of answers suggesting that the parasitic wasps would eat the crop itself once there were no more mealybugs.

QUESTION 02

02.1

Many answers were too vague to gain the mark for this question as neither 'competition' nor naming the type of competition were sufficient. The most common correct answers either referred to competition for food or occupying different niches.

02.2

This question proved surprisingly challenging with less than a third of students gaining the mark for identifying the type of selection and giving the reason. The question provided the mean body length and range, then gave the body lengths of the two populations; the data in the table clearly showed body lengths at each end of the range being favoured by the different populations. The correct use of a key term was also an issue with this question with answers using diversifying or disturbance instead of disruptive.

02.3

This was a fairly straightforward question asking for a definition of sympatric speciation. Many students gave clear concise answers and almost a quarter gained both marks. Answers that did not gain marking point one tended to miss out the second half of the statement and just said it was the development of new species. A number of answers were seen where students incorrectly stated that sympatric speciation occurred when there was a geographical barrier present.

02.4

This question required a description of the process of speciation and some good answers gaining all 4 marks were seen. Many answers lacked logical structure and sequencing. Students should be reminded to plan their answers to extended response questions to ensure that information is written in the correct order.

Students should be reminded that speciation has occurred when the two populations can no longer interbreed with each other producing fertile offspring; some answers did not make it clear that it was **interbreeding** leading to infertile offspring and just referred to **breeding.**

QUESTION 03

03.1

This question required students to **apply** their knowledge of aerobic respiration to explain the effect of methionine as a respiratory substrate. Some very good answers were seen showing good understanding; the question was a good discriminator with less than 10% of students scoring the full 3 marks.

When provided with an overview of the main stages of aerobic respiration as shown in Figure 3, students should spend some time looking at and understanding the diagram before attempting the questions.

03.2

Most students made good use of Figure 3 and applied their knowledge well. Occasionally, students just described the role of oxaloacetate rather than answering the question.

03.3

Very good application of knowledge was seen, but less than a quarter of students gained full marks for making clear links between enzyme inhibition and its effect on coenzymes, the electron transfer chain, and the use of oxygen as the final electron acceptor.

QUESTION 04

04.1

This question required students to explain why a genetic cross produced females with red eyes, males with red eyes, and males with white eyes in an approximately 2:1:1 ratio. Most students showed good understanding and clear genetic diagrams were drawn with genotypes clearly linked to phenotypes gaining over two thirds of students both marks. Some students lost the second marking point despite identifying parental genotypes and correctly drawing a diagram as they did not link the offspring genotypes and phenotypes together.

A number of students gave an incorrect parental genotype and did not get marking point 2 either as their offspring phenotypes included white-eyed females.

04.2

Most students showed good understanding of sex-linkage and how it can affect the expression of the genotype leading to over half of students scoring both marks for this question. The marking points required an idea of alleles being expressed, not just a vague reference to females having XX and males having XY chromosomes.

04.3

This was a well-answered question with almost two thirds of students scoring both marks for identifying the gametes, then giving the correct offspring genotypes and phenotypes.

Students who lost marking point 1 tended to identify the gametes as **gg** or **nn** for example, rather than **gn**. Some students lost marking point 2 for not matching the genotypes with their phenotypes.

For the parent with genotype **ggnn**, there was only one possible gamete: **gn**. Some students drew a Punnett square with four identical rows, each with the **gn** gamete, which was unnecessary.

04.4

Although some very good answers were seen with almost 10% of students scoring the full three marks, there were also many answers that showed little understanding of autosomal linkage, crossing over, and formation of recombinants.

Incorrect answers often referred to mutations, random mating, random fertilisation, or selection pressures. Some also suggested that the sample size was too small, or that correlation does not mean causation.

04.5

Good understanding and interpretation of chi-squared were seen, but almost 10% of students did not attempt the question. Some answers were still seen that referred to ‘results not due to chance’ or ‘results significant’ when it was essential to state that the <u>difference</u> between observed and expected was significant.

QUESTION 05

05.1

Good suggestions were seen for limitations of an unfamiliar method. Over 10% of students scored the maximum two marks for recognising that the size of beads may differ, and the number of algal cells may not be consistent. Common incorrect answers focused on control variables, for example, temperature and pH have not been controlled.

05.2

This question just required students to realise that the algae release carbon dioxide during respiration as well as taking it in for photosynthesis. Almost a quarter of students made this link with common incorrect answers suggesting that pH changed due to other factors,

05.3

Many students interpreted the term ‘clean’ to mean sterile and a lot of answers then went on to suggest that bacteria or fungi may compete with the algae. Marking point, one was most commonly awarded with fewer students realising that dirty glassware would affect the absorbance of the solution.

05.4

There was a noticeable improvement in graphing skills compared to previous series with almost a third of students gaining the full 3 marks. These graphs made good use of the grid provided, had linear scales and labelled axes, points were clearly plotted using a small cross, and a smooth curve of best fit was draw that plateaued at an absorbance of 0.71.

Occasionally, non-linear scales were seen so students did not gain marking point 1. The most common marking point to lose was the final one for the suitable line of best fit. Despite this instruction in the question, some students joined point-to-point using a ruler, or drew a straight line passing through some of the points rather than a curved line.

05.5

Having drawn a suitable graph in 05.4, the anomalous result was clearly identifiable, and most students could state that the point did not lie on the curve of best fit or did not follow the expected trend.

05.6

Only a fifth of students gained the mark for suggesting a suitable reason for the anomalous result. The reason had to specifically state why the result for absorbance was lower than expected, so answers that did not gain the mark were too vague including different number or algal cells rather than **fewer** cells, or apparatus not set up properly.

05.7

Many answers made a good link between either increased or decreased temperature and how it would affect production of organic molecules. Answers gaining one mark tended to just state how temperature affects enzymes but did not make the link to biomass production. Some answers incorrectly stated that low temperatures would denature enzymes or made vague reference to temperature *affecting* enzymes without specifying how.

05.8

Some excellent evaluations were seen with a definite improvement compared to previous exam series; a few of these answers were awarded the maximum 4 marks but included up to 6 correct marking points. It was less common to see answers listing generic phrases eg ‘correlation does not mean causation’, and there was good evidence of students looking at both the method and data carefully.

Although all marking points were awarded, 1, 2, 4, and 6 were the most common. Some students did not gain marking point 1 as they stated that a high concentration of carbon dioxide increased mean dry mass rather than recognising that it gave a **greater** increase than normal carbon dioxide concentrations. It was also good to see fewer students writing ‘results are not significant’ or ‘results are due to chance’ for marking point 2.

QUESTION 06

06.1

Over 40% of students gained the mark for correctly suggesting a management technique, with grazing and burning the most common answers. Answers that were too vague to gain the mark included preventing plant growth, clearing land, or preventing the climax community as these answers did not specify how this would be achieved.

06.2

Some good application of knowledge was seen in answers to this question with correct use of terminology and understanding of the management of succession. A sixth of students gained both marks. Occasional misuse of words was seen with ‘climate community’ and ‘climax communication’.

06.3

Some very good answers were seen to this question with many students recognising that no two species can occupy the same niche and some answers using the phrase ‘competitive exclusion principle’. Answers that did not gain the mark tended to make vague reference to competition without further explanation of its effect.

06.4

Most answered correctly and gained both marks for this calculation. A small number omitted one of the steps and only gained 1 mark.

06.5

Some very good evaluations were seen with students making good use of the information provided. It was good to see fewer comments like ‘correlation does not mean causation’ as rote learnt responses to ‘evaluate’ questions. All marking points were seen with points 1 and 8 the most commonly awarded for recognising the overall trend and the relevance of overlapping error bars. It was surprising that marking point 9 was infrequently awarded as the questions showed that measurements were only taken in two years.

The main reason for students losing marks in the question was incorrectly referring to ‘results due to chance’ or ‘results are significant’ rather than referring to the difference or the decrease.

QUESTION 07

07.1

A lot of clear, concise, well-written responses to this question were seen with over a third of students scoring the full 6 marks, and two thirds scoring 4 marks or more. Use of terminology was good, and most students had a sound understanding of the role of bacteria in the nitrogen cycle.

The most common errors were in marking points 2 and 4 when students referred to ammonium rather than ammonium **ions** or missed off the charges when writing out the chemical formula for ions eg NH_4 rather than NH_4^+.

07.2

Unfortunately, although many answers to this question started well by identifying the light-independent reaction of photosynthesis as one way that green plants could affect the concentration of carbon dioxide in the atmosphere, but then just described this process in a lot of detail and did not consider other ways that green plants could have an effect. Some answers also described the light-dependent reaction, which was irrelevant.

Vague references to photosynthesis or respiration lacked the level of detail expected at A-level for marking points 1 and 2.

07.3

Students are expected to know that, although the greenhouse effect is essential to life on Earth, carbon dioxide and methane enhance the effect bringing about climate change. Some good, clear answers were seen with over 10% of students gaining 4 or 5 marks. The marking points most commonly awarded were 1 and 2 for the effects of deforestation and burning fossil fuels on levels of carbon dioxide in the atmosphere.

Occasionally, students started writing about the effects of other human activities such as release of acidic gases, or hunting; these were irrelevant as the question specifically focused on climate change. Many students made inappropriate references to the effect of carbon dioxide on the ozone layer.